Project Initialization and Planning Phase

Date	June 28, 2026
Team ID	
Project Title	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This project focuses on optimizing agricultural production using machine learning techniques. It analyzes soil nutrients and environmental conditions to recommend the most suitable crops and assess agricultural productivity.

The solution reduces guesswork in farming, improves yield efficiency, supports sustainable practices, and enables data-driven agricultural decision-making.

Project Overview	
Objective	Develop an intelligent machine learning system that recommends the most suitable crop based on soil nutrients (N, P, K) and environmental conditions such as temperature, humidity, pH, and rainfall.
Scope	The project includes: • Agricultural data collection • Data preprocessing and feature selection • Model training and evaluation • Crop recommendation system • Deployment using Flask for real-time predictions Multiple machine learning algorithms may be tested to identify the best-performing model.
Problem Statement	
Description	Farmers often rely on traditional knowledge or trial-and-error methods to decide which crops to cultivate. This can lead to poor crop selection, low productivity, and inefficient use of resources due to lack of data-driven insights.
Impact	• Reduced agricultural productivity • Inefficient use of fertilizers and water • Increased financial risk for farmers • Environmental degradation due to improper crop selection
Proposed Solution	
Approach	Agricultural datasets containing soil nutrients (N, P, K) and environmental factors (temperature, humidity, pH, rainfall) are preprocessed and used to train machine learning models such as: • Decision Tree • Random Forest • Logistic Regression The best-performing model is integrated into a Flask web application to provide real-time crop recommendations.

Key Features	• Smart Crop Recommendation System • Data-Driven Agricultural Decision Support • Real-Time Prediction using Web Application • User-Friendly Interface for Farmers • Environmental Suitability Insights • Efficient Resource Utilization Support • Scalable and Practical System Design

Resource Requirements

Resource Type	Description	Specification/ Allocation
Hardware		
Computing Resources	CPU for model training and deployment	Intel Core i5/i7 Processor (4+ Cores)
Memory	RAM for processing datasets	Minimum 8 GB (16 GB recommended)
Storage	Dataset and model storage	512 GB SSD / 1 TB HDD
Software		
Frameworks	Python Web Framework	Flask
Libraries	Machine Learning and Data Processing Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost
Development Tools	IDE	Jupyter Notebook, Visual Studio Code, PyCharm
Data		
Dataset	Agricultural Crop Recommendation Dataset	Collected from publicly available sources (e.g., Kaggle), used for training ML model
Format	CSV	Structured tabular format with rows (records) and columns (features)
Attributes	Soil and environmental parameters	See detailed attribute specifications below
Size	~2,000–5,000 records	Sufficient for training basic ML models with good accuracy